

MountMonitor

User Manual

Real-time telescope mount monitoring
for astrophotography

Version 1.33.0

Theory · Installation · Use · Reference

After MountMonitor v3.37 by Nicolàs de Hilster
Python/PyQt6 rewrite

MIT | github.com/ARP273-ROSE/MountMonitor

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Part I

Understanding

Chapter 1

Celestial coordinates

1.1 The celestial sphere

Seen from the ground, the night sky behaves like a sphere turning from east to west about an axis through the celestial poles. Stars lie at wildly different distances, but their position on that fictitious sphere is all one needs to find and follow them.

The Earth completes one turn in **23 h 56 min 4 s** — the *sidereal day*, shorter than the 24 hours of the solar day. That, and not the clock, is the rate a mount must turn at.

1.2 Right ascension

Right ascension (RA) is the celestial equivalent of longitude. It is counted in **hours** rather than degrees, from 0 h to 24 h, starting at the vernal point.

Formula

$$1\,\mathrm{h} = 15^\circ \qquad 1\,\mathrm{min} = 15' \qquad 1\,\mathrm{s} = 15''$$

So RA = 02 h 44 min 08 s is $2 \times 15^\circ + 44 \times 0.25^\circ + 8 \times \frac{15}{3600}^\circ = 41.03^\circ$.

Information

RA is the critical coordinate for tracking: a one-second timing error is worth 15 arcseconds on the sky at the celestial equator. At 1.6″/pixel that is about ten pixels of trailing.

1.3 Declination

Declination (DEC) is the equivalent of latitude: the angular distance north or south of the celestial equator, from -90° to $+90^\circ$. On a correctly aligned mount the declination axis does not move while tracking.

1.4 Hour angle

The **hour angle** measures how far an object stands from the local meridian. It grows continuously as the Earth turns.

Formula

$$\mathrm{HA} = \text{local sidereal time} - \mathrm{RA}$$

An RA tolerance can be expressed in arcseconds or in seconds of time. The conversion depends on declination:

$$\text{seconds of time} = \frac{\text{arcseconds}}{15\cos(\mathrm{DEC})}$$

Tip

Near the celestial pole an RA error barely moves the star on the sensor: the cosine of the declination damps it. Circumpolar objects are therefore more forgiving of RA tracking errors — but not of declination ones.

Chapter 2

Equatorial mounts

2.1 Principle

An **equatorial mount** cancels the Earth's rotation by turning the tube about an axis parallel to the Earth's own: the polar axis.

- **Polar axis (RA)**: turns at the sidereal rate.
- **Declination axis**: still while tracking, in theory.

Formula

$$\omega = \frac{1\,296\,000''}{86\,164.09\,\mathrm{s}} \approx 15.041\,\text{arcseconds per second}$$

Any departure from that rate, and any misalignment of the polar axis, becomes a drift the image records.

2.2 10Micron mounts

10Micron mounts (GM1000 HPS and above) carry **absolute encoders**: they know where they point even after a power cut, where a relative encoder loses it. To that they add a worm drive, periodic-error correction and, above all, an internal **pointing model** built from several dozen stars.

That model is what makes **unguided** imaging possible: the mount does not merely turn at the sidereal rate, it applies corrected rates that absorb flexure, refraction and the residual polar misalignment.

Warning

A direct consequence, often misunderstood: on a modelled mount the **polar alignment error lives inside the model** and is already compensated by the tracking rates. A residual declination drift therefore does not call for the azimuth screw, but for **more model points near the target**.

These mounts speak an **extended LX200**: beyond the standard commands they expose raw axis positions, applied rates, model state and environmental data. MountMonitor uses those extensions where available.

Chapter 3

What degrades tracking

No mount tracks perfectly. The causes overlap, and each leaves a different signature in the record.

3.1 Periodic error

A worm is never perfectly round nor perfectly centred. Its defect repeats with every turn and produces an oscillation of fixed period — a few minutes, most often. That is exactly what a Fourier analysis can pull out (chapter 5).

3.2 Polar alignment

If the polar axis does not point exactly at the celestial pole, the star drifts slowly in declination, the faster the further one is from the meridian. On a mount without a model the answer is mechanical. On a modelled mount, see the warning in the previous chapter.

3.3 Atmospheric refraction

The atmosphere lifts objects the more they are low: some $34'$ at the horizon, $1'$ at $45°$ of altitude. Refraction therefore changes through the night, and its drift is easily mistaken for a misalignment.

3.4 Mechanical disturbance and vibration

Wind, a cable pulling, ground movement, a meridian flip: so many one-off jolts, none of them periodic. They show as isolated spikes, and that is what tells them apart.

3.5 Commanded moves

These are not errors at all. Between exposures the sequencer deliberately moves the mount — **dither** to decorrelate noise, **re-centering** when the target has slipped — and the mount *stays* at its new position. Confusing them with tracking error is the costliest mistake one can make reading a record, and chapter 4 is devoted to it.

Chapter 4

Reading the numbers

Warning

This is the most important chapter in the manual. A tracking record is read wrongly with disconcerting ease, and the figure that comes out is then wrong by a factor of several hundred.

4.1 The signal is a staircase

Here is what a night of unguided imaging actually records: the mount holds its position to a fraction of an arcsecond, then the sequencer dithers it, and it holds its new position just as well. The trace is not a noisy line about a mean: it is a **staircase**.

A standard deviation taken over the whole night then measures the height of the steps, not the quality of the tracking. And removing a straight line — what a classic detrending does — does not remove a staircase.

Information

On a real eight-and-a-half hour session the raw combined RMS came out at **86.3″**, a damning verdict. The mount was in fact holding each position to **0.18″**. The two figures differ by a factor of 480.

4.2 Jitter

Jitter is the spread the mount shows *while* it holds a position. It is the only figure that smears a star on the sensor, and the one MountMonitor bases its rating on.

It is measured between repositionings, never about a straight line. The program cuts the record at the steps, drops two samples after each — the move itself is not tracking — and takes the median of the per-step spreads.

Tip

This approach has a property worth noting: a sequence that dithers on every frame does not come out worse than one that never does. The figure describes the mount, not the sequencer.

4.3 Commanded moves

They are counted and classified separately:

Class	What it is
Dither	A small move between exposures, to decorrelate noise.
Re-centering	A larger move, bringing the target back where it belongs.
Unsettled	The spread that follows stays several times the usual value. That is the one worth looking at.

Warning

A threshold crossing is not a move. A 70″ re-centering takes several samples to complete and trips the threshold at each one: one night read as 98 crossings for 53 actual moves, with a median amplitude ten times too small. MountMonitor groups crossings less than ten seconds apart.

4.4 Slow drift

It is quoted per hour and per exposure. On an unguided mount every dither cancels it: what matters is not the drift over the night, but the displacement it produces *during one exposure*.

Formula

$$\text{displacement over one exposure} = \text{drift } (''/\text{h}) \times \frac{\text{exposure (s)}}{3600}$$

A drift of $-16.8''/\text{h}$ therefore moves the star by $0.84''$ over a 180 s exposure.

4.5 Running standard deviation

Computed live over a 60 to 900 second window, it serves to see *when* something happened, not to grade the night. After a target change, an acquisition gap or a repositioning, its window straddles the move and describes that rather than the tracking: those values are excluded from the report.

4.6 The rating

It rests on the combined jitter, with thresholds suited to the type of mount. An unguided 10Micron is judged on wider thresholds than a guided mount, because the orders of magnitude are not the same.

Rating	Guided mount	Unguided precision
Excellent	$< 0.5''$	$< 2.0''$
Good	$< 1.5''$	$< 4.0''$
Fair	$< 3.0''$	$< 8.0''$

Tip

Always relate the rating to your sampling. At 1.6″/pixel, a 0.5″ jitter shakes the star by a third of a pixel: invisible on the final image.

Chapter 5

Fourier analysis

5.1 What it is for

A Fourier transform breaks a signal into a sum of sinusoids. On a tracking record it brings out what *repeats* — and a worm periodic error repeats by construction.

MountMonitor looks in the **2 to 15 minute** band, where the worm periods of common mounts fall.

5.2 Two precautions

- The transform runs on **detrended** deviations. A linear drift is not periodic, but over a finite window it produces a large low-frequency component that would hide what is being looked for.
- The **Nyquist limit** requires sampling at least twice per period. At 2 Hz, periods below one second are out of reach — no real inconvenience, no mechanical error lives there.

5.3 Reading a spectrum

A sharp peak at a stable period marks a periodic error. A flat spectrum means there is no detectable one — the case of 10Micron mounts after correction. Broad, wandering peaks are noise, usually wind.

Part II

Using

Chapter 6

Installation

6.1 Supported systems

The published packages carry their own Python and their own Qt: nothing else needs installing. What they do require is a recent enough system, because the Qt binaries inside them require it.

System	Minimum	Architecture
Windows	Windows 10 (1809)	x64
macOS	macOS 11 Big Sur	Apple Silicon *and* Intel
Linux	glibc 2.28 — Debian 11, Ubuntu 20.04, RHEL 8	x86_64
Linux ARM	glibc 2.39 — Ubuntu 24.04	arm64

Warning

Pick the macOS package that matches your machine: `-macos-arm64.dmg` for an Apple Silicon Mac, `-macos-x86_64.dmg` for an Intel one. The bundled Python is architecture-specific even though Qt itself is universal: the wrong package will not start at all.

6.2 Installing

On Windows the installer handles everything. On macOS, drag the application into the Applications folder. On Linux, `installer.sh` installs it into the user profile, with neither `sudo` nor a package manager.

Updates are then checked quietly at every start and offered when there are any.

6.3 A first run without a mount

Simulation mode lets you walk through everything with no hardware:

```
MountMonitor --sim-all
```

Here is the window after half a minute of simulation: at the top the right ascension and declination deviations with their tolerances, then the PC–mount clock difference and the seismometer; on the right the mount's state, its coordinates and the session log.


File View Reset Edit Language Help
Disconnect
ARMED — waiting for tracking
FFT
Preferences
RIGHT ASCENSION
+0.36" σ=0.17"
Deviation (")
Time (s)
DECLINATION
+0.10" σ=0.23"
Deviation (")
Time (s)
TIME — PC vs Mount Clock
PC-Mount: +370.1ms Drift: +16.58ms/min PC loop: 500.1ms Mt loop: 1000.0ms
PC-Mount
PC loop
Mount loop
PC-NTP
Time diff (ms)
Time (s)
SEISMIC — Vibrations
Amplitude
Time (s)
The mount is tracking
RA: 02:44:08.61 Δ +0.36" σ 0.18"
DEC: +61:10:17.0 Δ +0.10" σ 0.24"
2.0 Hz | 59 samples
15:47:15 Temp: 4.9°C Press: 1013mbar Int: 15.0°C Flip: 180min
15:47:15 Mount: MountMonitor Simulated Mount | Firmware: SIM-1.0 | Pier: West
15:47:15 Mount parked: tracking checks skipped.
15:47:15 Refraction: continuously updating ✓
15:47:15 Astronomical night 19:16 → 05:01 (9 h 44)
15:47:15 Sunset 17:41 | Naut. twilight 18:43 | Sunrise 06:37
15:47:15 Site : +43:55:54.1 +05:42:43.9 elev. 650 m
15:47:15 Az: 341.3° Alt: 24.5°
15:47:15 Pier side : West
15:47:15 ID : SIM-00000000000000000001
15:47:15 Firmware : SIM-1.0

Chapter 7

Settings

7.1 Connection

Protocol	When to use it
LX200 TCP/IP	A 10Micron on the network. Port 3492 by default.
LX200 serial	RS-232 link, 9600 8N1 as shipped on 10Micron mounts.
ASCOM	Windows only, through the manufacturer's driver.

7.2 The observing site

Warning

Fill in **latitude, longitude and elevation** in the preferences. The mount is asked first (`:Gt#` and `:Gg#`), but many setups never pushed their site to it and it then answers nothing. Without a site the night ephemeris cannot be computed.
Longitude is entered **positive east**. The LX200 protocol counts the other way; the conversion is done for you when the mount answers.

7.3 Tolerances and running window

The tolerance is an alert threshold you set yourself. A simple rule: take your sampling in arcseconds per pixel. A tolerance equal to the sampling is strict, twice that is comfortable.

The running standard deviation window is set between 60 and 900 seconds. Sixty reacts fast, nine hundred smooths a long session.

7.4 Connecting automatically at launch

On an observatory PC that starts on its own, MountMonitor can connect to the mount as soon as it opens, without a click. Two equivalent ways:

- tick **Connect to the mount when MountMonitor opens** in the Preferences (recording tab);
- start the program with the `-connect` option, for instance from a scheduled task.

When the PC starts, the mount's control box has often not finished booting: while the mount does not answer, MountMonitor tries again every 30 seconds (setting `connect_retry_s`, 5 s minimum) until the first successful connection, and says so in the session log. A click on *Connect* or *Disconnect* stops the attempts. No dialog opens during these attempts: with ASCOM the driver must already be chosen; on a serial link the port must already be set. With `-connect`, the offer to report a previous crash is kept for the next launch, so that it does not hold the window.

Once connected, recording follows the settings below: it is armed and starts when the mount starts tracking.

7.5 Arming the logger

Three settings govern the record button. All three are **on by default**.

Start recording when the mount starts tracking. The button then *arms* the logger instead of starting it: nothing is written until the mount tracks. A mount connected at noon for a night beginning at 19:00 no longer records seven hours of nothing. A second click disarms.

Suspend when the mount stops tracking, with a two-minute grace delay. The delay exists for a reason: a stop may be transient — a meridian flip, an autofocus run, a slew between targets — and cutting on the first sample would punch a hole at every dither.

End the night at sunrise. A night is one unit: if the mount parks at 02:00 and resumes at 03:00, recording suspends and resumes *in the same file*. Only daylight closes the session and triggers the report — and only once the Sun has actually been below the horizon during the session, so that a daytime trial does not close itself.

Chapter 8

A night out

8.1 How it goes

1. Connect the mount and check the status turns to tracking.
2. Arm the logger — or start it, if arming is switched off.
3. Leave it running. The report will write itself at the end.

8.2 What to watch live

- **The running standard deviation**, rather than the instantaneous deviation: it says whether the mount is holding its position.
- **Tolerance alerts**, which flag an overshoot.
- **The dew-point margin**, if an environment sensor is connected.

> **Tip**
>
> An isolated spike is almost never the mount: it is a gust, a cable, a footstep too close to the pier. What matters is the level *between* the spikes.

8.3 Night ephemeris

Computed from the site, it gives sunset, nautical twilight, astronomical night and sunrise — and above all how much of the dark night the session actually covered. A night started an hour late loses an hour that cannot be recovered, and that is more useful read beside the tracking figures than after the fact.

Chapter 9

The files

9.1 What a session writes

Extension	Contents
`.dat`	The samples: time, RA, DEC, standard deviations, status.
`.dti`	PC / mount synchronisation measurements.
`.fft`	Spectra recorded during the session.
`.env`	Temperature, pressure, humidity, when the mount reports them.
`.log`	Events: connections, slews, alerts, suspensions.

The `.dat` header carries the site and the time offset. That is what allows twilights to be recomputed on replay, long after the mount has been unplugged.

9.2 The night report

It is written automatically at the end of a session, next to the `.dat`, as **one file per language**. Nothing has to be asked for.

9.3 Replay

File → **Open log** reads a finished session and rebuilds the whole analysis: per-target statistics, commanded moves, drift, spectra, ephemeris. The language tabs show the same report in all three languages.

Chapter 10

When something is wrong

Symptom	Where to look
No network connection	Check the address and port 3492, and that nothing else already holds the session — a 10Micron accepts only one.
Serial link silent	Both ends must agree on the speed. 10Micron mounts ship at 9600 baud, and the mount's own menu can change it.
Declination drift	On a modelled mount, add model points near the target. The azimuth screw is not the answer.
Poor rating, quiet mount	Read chapter 4: it is often the reading that is wrong, not the mount.
Absurd running STDEV	Normal just after a slew or a repositioning: its window straddles the move. The report excludes those values.
No ephemeris	The site is not set. Preferences, site section.

Chapter 11

Reference

11.1 Shortcuts

Keys	Action
`F1`	Help
`Ctrl+,`	Preferences
`Ctrl+O`	Open a log
`Ctrl+Q`	Quit

11.2 Language

The **Language** menu offers English, French and Dutch, plus automatic detection from the system. Each entry is written in its own language: someone who ends up in the wrong one cannot read the current one to find their way out. A change takes effect at the next start.

11.3 Problem reports

If the program crashes, freezes or fails to start, it can report that by itself. You are asked once, and nothing else is ever sent: not your tracking data, not your file names, not your user name. All paths are anonymised.

In the **Help** menu, **Report a problem…** sends a message written by you to the same collection point, and the **Send crash reports automatically** box withdraws or gives back consent at any time. **Report a Bug on GitHub** opens a pre-filled issue on the project's public page instead.

11.4 Version history

The detail of every change, with the measured figures behind it, is in the `CHANGELOG.md` file shipped with the program.

11.5 Origin

MountMonitor is an independent Python and PyQt6 rewrite of MountMonitor v3.37 by **Nicolàs de Hilster** (2018–2021), written in Java. No code from the original program appears in it. The MIT licence of the present program covers this rewrite only; the original work remains subject to its author's terms.